Predictors of Premature Mortality at the County Level in the United States

County-Level Public Health Analysis

07 July 2026

Contents

1 Introduction

County-level health outcomes vary widely across the United States, and the factors behind that variation matter for how scarce public-health resources are allocated. This report answers the question: **what predicts better health outcomes at the county level in the United States?** Following the project brief, the outcome is *premature age-adjusted mortality* (deaths before age 75, age-standardised, per 100,000 population) from the 2018 County Health Rankings. Because lower mortality represents better health, the aim is to identify which county-level social, behavioural, healthcare-access, and drug-environment variables are most strongly associated with better health outcomes.

Premature mortality is a useful summary of population health because it weights the deaths that are, in principle, most avoidable through prevention and timely care. Measuring it by county matters for policy, because much U.S. health funding and programme delivery is administered locally, so identifying which county characteristics travel with worse outcomes helps target limited resources. The variation is also large enough to be worth explaining: across the analysis sample, premature mortality ranges from about 133 to 940 deaths per 100,000.

This study argues that premature mortality is shaped jointly by four domains: health behaviour, socioeconomic disadvantage, healthcare access, and the local drug environment. It tests this by building a multivariate linear regression model with six predictors and one externally sourced variable (CDC opioid dispensing). Previewing the main findings: socioeconomic disadvantage (child poverty) and behavioural risk (smoking) are the strongest predictors, opioid dispensing adds independent explanatory power, and — contrary to expectation — graduation rate is *not* protective once the other variables are controlled, a pattern this report interprets as a confounding/suppression effect.

2 Hypotheses

- **General hypothesis:** counties with healthier behavioural profiles, lower socioeconomic disadvantage, better healthcare access, and a less harmful drug environment have lower premature mortality.

The specific hypotheses below state both the expected *direction* and, drawing on the literature reviewed in the next section, the expected *strength* of each association.

- **H1 (smoking):** higher adult smoking rate → higher premature mortality; expected to be one of the strongest predictors, as smoking is the leading preventable cause of death.
- **H2 (obesity):** higher adult obesity rate → higher premature mortality; expected to be a moderate predictor, since the obesity–mortality association is smaller and more contested than that for smoking.
- **H3 (uninsured):** higher uninsured rate → higher premature mortality; expected to be a moderate predictor.
- **H4 (education):** higher high-school graduation rate → *lower* premature mortality; expected to be protective but largely *indirect*, so its independent effect may be weak once behaviour and poverty are held constant.
- **H5 (child poverty):** higher child poverty rate → higher premature mortality; expected to be among the strongest predictors, as socioeconomic disadvantage is a foundational determinant of health.
- **H6 (opioids):** higher opioid dispensing rate → higher premature mortality; expected to be positive but secondary to the socioeconomic and behavioural factors.

3 Literature and theory

County premature mortality varies widely across the United States, and that variation tracks social conditions far more than chance. This study draws on the *social determinants of health* framework, which links health to the circumstances in which people live, work, and age. In a multicohort study of 1.7 million people, Stringhini et al. (2017) show that low socioeconomic status predicts premature mortality about as strongly as established behavioural risk factors, and that these disadvantages accumulate over the life course. Building on this framework, the present study treats county premature mortality as jointly shaped by four established domains — socioeconomic disadvantage, educational attainment, healthcare access, and behavioural risk — with a fifth contextual factor, the local drug environment, added for this project.

Socioeconomic disadvantage is the foundation. Childhood poverty is associated with worse health across the life course and elevated mortality risk (Wise, 2016), in part through early-life biological embedding that raises later cardiovascular risk (Galobardes et al., 2006). Counties with higher child poverty are therefore expected to show higher premature mortality (H5, positive).

Education is consistently one of the strongest predictors of longevity. A global systematic review finds that additional schooling lowers adult mortality (IHME-CHAIN Collaborators, 2024), a gradient confirmed in U.S. cohorts (Kaplan et al., 2015). Crucially, much of education's protective effect operates *indirectly*, through the behaviours and material conditions it shapes: Puka et al. (2022) show that lifestyle risk factors such as smoking explain a substantial share of the education–mortality gradient. This mediated structure is central to the analysis, because it implies that the graduation-rate coefficient may weaken or even change sign once smoking and poverty are held constant (H4, expected negative but plausibly mediated).

Access to care matters independently of behaviour. Systematic reviews show that health insurance increases the use of effective care and improves outcomes (Freeman et al., 2008), and that uninsured U.S. adults face higher mortality (Wilper et al., 2009; Woolhandler & Himmelstein, 2017). This evidence is weaker than it first appears, however: it is largely observational and open to reverse causation and selection — sicker and poorer people are also more likely to be uninsured — so the independent effect of insurance is harder to isolate than that of a behavioural risk. A higher uninsured rate is therefore expected to predict higher mortality, though plausibly only modestly (H3, positive).

Behaviour is the most direct pathway of all. Smoking remains the leading preventable cause of death in the United States (U.S. Department of Health and Human Services, 2014), is strongly associated with all-cause mortality (Lariscy et al., 2018), and its harms are at least partly reversible after cessation (Mons et al., 2015). Obesity likewise raises all-cause mortality risk across large meta-analyses (Flegal et al., 2013; Aune et al., 2016; Jayedi et al., 2022), though the size of this association is contested: estimates vary with how body mass is categorised, and BMI is an imperfect proxy for adiposity (Flegal et al., 2013). Both smoking and obesity are therefore expected to be positive predictors, with obesity the weaker of the two (H1, H2).

Finally, this project adds the local drug environment. County-level drug mortality is closely tied to the supply of prescription opioids and to the economic distress of place (Monnat, 2018; Dasgupta et al., 2018). This evidence is itself largely ecological — established across places rather than individuals — so it justifies including the opioid dispensing rate as an external predictor without, on its own, establishing individual-level risk (H6, positive).

These domains do not act in isolation. Most of education's protective effect is expected to run through behaviour and material conditions rather than directly, which is why the graduation-rate coefficient may weaken once those factors are held constant. Because these determinants are measured at the county rather than the individual level, the analysis speaks to ecological association rather than individual causation, and aggregate correlations need not hold within individuals (Robinson, 1950).

4 Data and methodology

4.1 Data sources

Table 1: Data sources and their role in the analysis.

Dataset	Provider	Role	Merge
County Health Rankings 2018	Robert Wood Johnson Foundation & Univ. of Wisconsin Population Health Institute	Outcome + 5 predictors	FIPS county code

Dataset	Provider	Role	Merge
CDC opioid dispensing, 2016	U.S. Centers for Disease Control and Prevention	External predictor (opioid_rate_2016)	FIPS county code

Table 1 summarises the two datasets and their roles. The provided dataset is the 2018 County Health Rankings (RWJF & University of Wisconsin Population Health Institute, 2018). The outcome, *premature age-adjusted mortality*, is taken from the Additional Measure Data sheet. Five predictors — adult smoking, adult obesity, uninsured rate, high-school graduation rate, and children in poverty — are drawn from the Ranked Measure Data sheet. As required, an additional external dataset is merged in: the CDC county-level opioid dispensing rate for 2016 (Centers for Disease Control and Prevention, 2017), joined on the five-digit FIPS county code.

The County Health Rankings are reasonably reliable because their component measures draw from established sources such as the CDC, the Census Bureau's American Community Survey, and the National Center for Education Statistics. Some measures are modelled estimates with confidence intervals, so county values carry sampling uncertainty. Bias mitigation is limited but explicit: counties are merged by FIPS rather than names, state roll-ups are filtered out, missingness is reported before modelling, and the 2016 opioid rate is treated as a lagged exposure rather than a contemporaneous measure.

4.2 Variables and descriptive statistics

Table 2: Descriptive statistics, complete-case analysis sample.

Variable	N	Mean	SD	Min	Median	Max
Premature age-adjusted mortality (per 100,000)	2611	403.3	106.1	133.0	391.9	940.0
Adult smoking (%)	2611	18.0	3.5	6.7	17.8	31.7
Adult obesity (%)	2611	31.6	4.5	12.8	31.8	47.8
Uninsured (%)	2611	11.7	4.9	2.1	11.3	33.0
High-school graduation rate (%)	2611	86.1	8.0	30.1	87.5	100.0
Children in poverty (%)	2611	22.5	9.1	2.9	21.8	66.3
Opioid dispensing rate, 2016 (per 100)	2611	79.9	39.8	0.0	76.0	399.9

Table 2 reports the mean, spread, and range of each variable across the complete-case sample; the predictors are measured on comparable percentage or rate scales, while the opioid dispensing rate shows the widest dispersion.

4.3 Missing data

Table 3: Missing values by variable (of 3142 counties). Complete-case N = 2611.

Variable	Missing
Premature age-adjusted mortality (per 100,000)	65
Adult smoking (%)	0
Adult obesity (%)	0
Uninsured (%)	1
High-school graduation rate (%)	470
Children in poverty (%)	1
Opioid dispensing rate, 2016 (per 100)	180

Listwise deletion is used, giving a complete-case sample of 2611 counties out of 3142. As Table 3 shows, most missingness is driven by the graduation rate (470 counties) and the opioid dispensing rate (180 counties).

Listwise deletion is a pragmatic but imperfect choice: it gives a transparent regression sample, but may under-represent smaller or more rural counties because missingness is concentrated in graduation and opioid data. Imputation is not used because assuming those counties resemble reporting counties is untestable; this representativeness issue is revisited in the limitations.

4.4 Method

No log transformation or recoding is applied in the main model because the selected variables are already interpretable percentages or rates. Coefficients are kept in their original units for substantive interpretation, while standardised coefficients are used only as a supplementary way to compare relative predictor strength.

The outcome is continuous, so the analysis uses **multiple linear regression (ordinary least squares)** with the six predictors entered simultaneously. OLS is chosen because it directly estimates the partial association of each predictor with mortality, holding the others constant, and the coefficients are interpretable as a change in deaths per 100,000. In the full model, each predictor also acts as a control variable for the others. The model is:

$$\text{Mortality}_i = \beta_0 + \beta_1 \text{Smoking}_i + \beta_2 \text{Obesity}_i + \beta_3 \text{Uninsured}_i + \beta_4 \text{Graduation}_i + \beta_5 \text{ChildPoverty}_i + \beta_6 \text{Opioid}_i + \varepsilon_i$$

Three modelling assumptions are examined below: linearity and constant error variance through a residuals-versus-fitted plot, and multicollinearity through variance inflation factors. As a method, OLS is well suited to this task — it yields transparent, directly interpretable partial coefficients — but it also assumes a linear, additive specification with independent errors, so it cannot capture spatial dependence between neighbouring counties or any non-linear effects. One feature of the design cannot be addressed by diagnostics. Because every variable is measured for the county as a whole, the coefficients describe differences between places rather than between people, and reading a county-level association as an individual-level effect would be an ecological fallacy; the interpretation throughout is framed accordingly.

5 Results

5.1 Exploratory data analysis

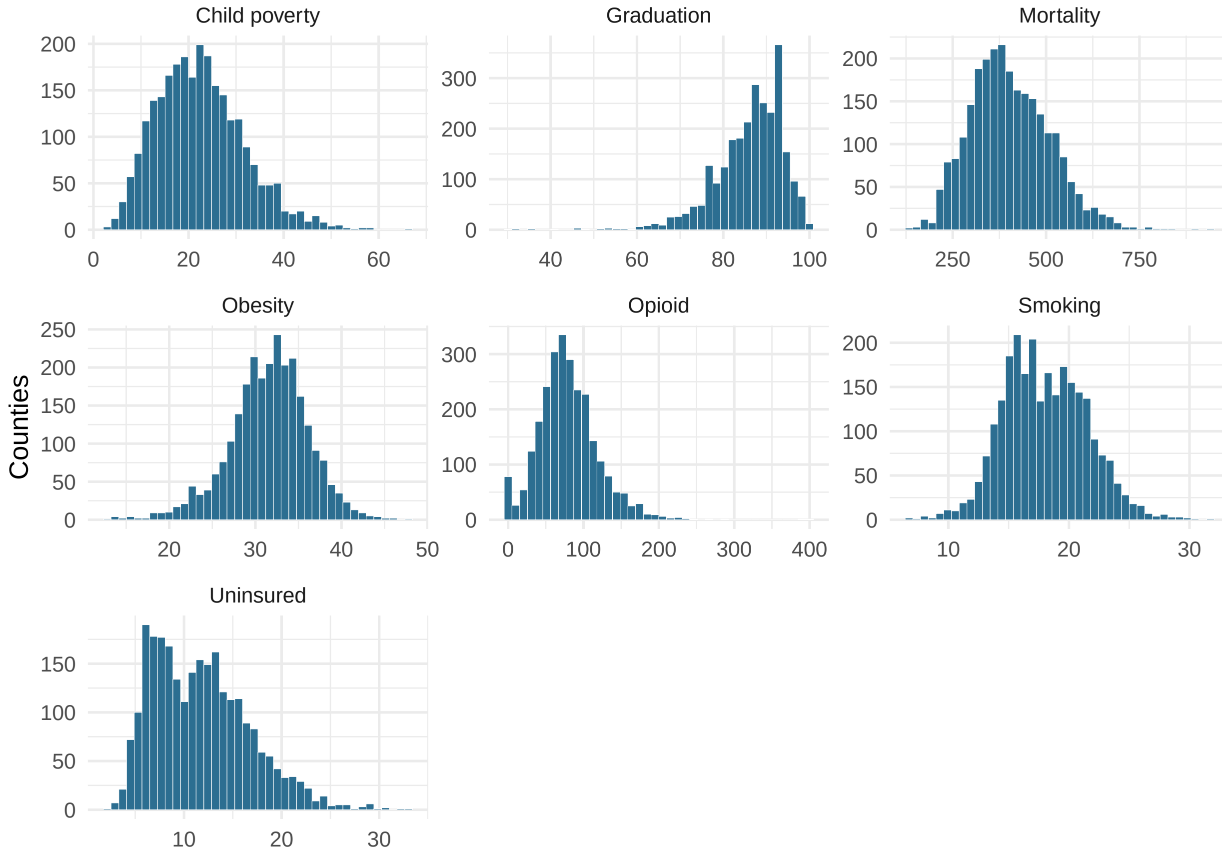

Figure 1: Distributions of the outcome and predictors across counties.

Figure 2: Pearson correlations among the outcome and predictors.

The outcome ranges from 133 to 940 premature deaths per 100,000 (mean 403). Bivariately (Figure 2), child poverty (r = 0.76) and smoking (r = 0.73) are the strongest correlates; graduation rate is only weakly (and negatively) associated (r = -0.10).

Two features of the data shape what follows. First, the predictors are themselves correlated — adult smoking and obesity move together across counties (r = 0.63), as do poverty and several behavioural measures — so the partial coefficients from the full model can differ from these simple correlations once the variables are entered together. Second, as Figure 1 shows, most variables are roughly symmetric, but the opioid dispensing rate is right-skewed, with a long tail of high-dispensing counties. The strong bivariate links from child poverty and smoking to mortality already foreshadow their prominence in the regression.

5.2 Regression model

Table 4: OLS regression predicting premature age-adjusted mortality. Coefficients are unstandardised; 95 per cent confidence intervals in brackets. N = 2611; R-squared = 0.732; adjusted R-squared = 0.731.

Predictor	b	SE	95% CI	t	p
(Intercept)	-69.310	14.138	[-97.03, -41.59]	-4.90	<.001
Adult smoking (%)	8.280	0.490	[7.32, 9.24]	16.91	<.001
Adult obesity (%)	3.678	0.310	[3.07, 4.29]	11.88	<.001
Uninsured (%)	2.388	0.253	[1.89, 2.88]	9.44	<.001
High-school graduation rate (%)	0.338	0.140	[0.06, 0.61]	2.41	0.016
Children in poverty (%)	4.656	0.187	[4.29, 5.02]	24.92	<.001
Opioid dispensing rate, 2016 (per 100)	0.570	0.029	[0.51, 0.63]	19.40	<.001

The full model (Table 4) explains 73.2% of the variation in county premature mortality. Five of six predictors carry the expected positive sign and are significant at $p < .001$: smoking, obesity, uninsured rate, child poverty, and opioid dispensing. The largest variance inflation factor is 2.48 for Children in poverty (%), well below the common threshold of 5, so multicollinearity does not indicate a serious threat to these estimates.

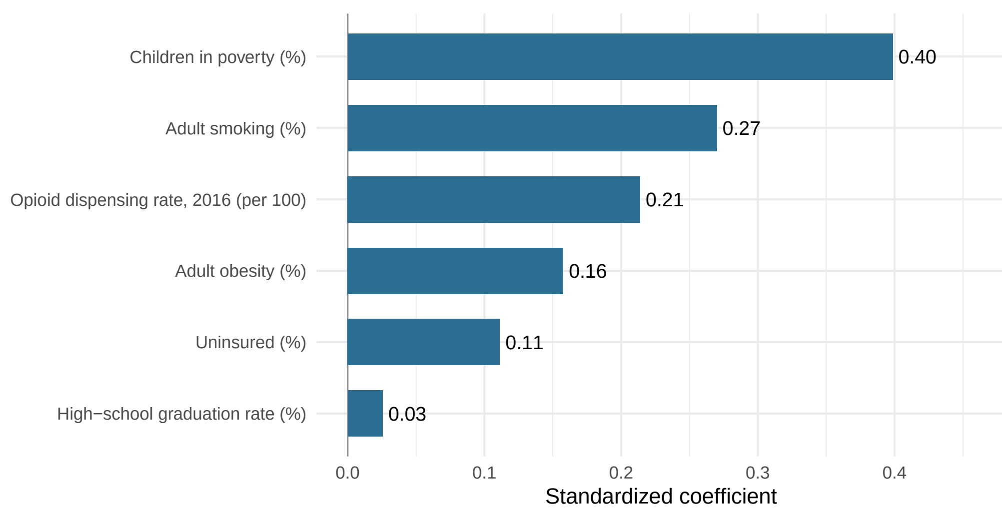

Figure 3: Standardized coefficients from the OLS model.

Because the predictors are measured on different scales, Figure 3 compares their relative strength using standardised coefficients. This confirms that child poverty is the strongest predictor, followed by smoking and opioid dispensing. Obesity and uninsured rate remain positive but smaller, while graduation rate is close to zero in practical terms.

The exception is **graduation rate**: H4 predicted a negative coefficient, but the estimate is small and *positive* (b = 0.34, p = 0.016), despite a negative bivariate correlation.

Table 5: Graduation-rate coefficient (unstandardised) before and after controlling for the other five predictors.

Model	b	SE	p
Model 1: graduation rate only	-1.349	0.258	$<.001$
Model 2: full model (6 predictors)	0.338	0.140	0.016

Table 5 makes this explicit: on its own, graduation rate has a negative coefficient, but it loses that once the other five predictors are controlled. This sign reversal does not mean education is harmful; it reflects *confounding* and a *suppression* effect rather than a real property of education. Considered on its own, graduation rate is negatively correlated with premature mortality (r = -0.10): counties that graduate more of their students do tend to have fewer premature deaths. That bivariate relationship, however, largely reflects the characteristics of counties with higher graduation rates — such counties also smoke less, are less obese, and have lower child poverty. Once smoking, obesity, uninsured status, child poverty, and opioid dispensing are entered as *control variables*, the graduation coefficient becomes a *conditional association*: the residual relationship that survives after holding those other determinants constant. Almost none does, which is exactly what the literature anticipates — education's protective influence operates largely *through* behaviour and material conditions rather than directly (Puka et al., 2022). Holding those pathways constant leaves little of the original negative signal — what remains is a small positive partial association. Because the design is cross-sectional and ecological, this is offered as an interpretation *consistent with* the data, not as evidence of causal mediation, and the county-level pattern should not be read as an individual-level effect (Robinson, 1950).

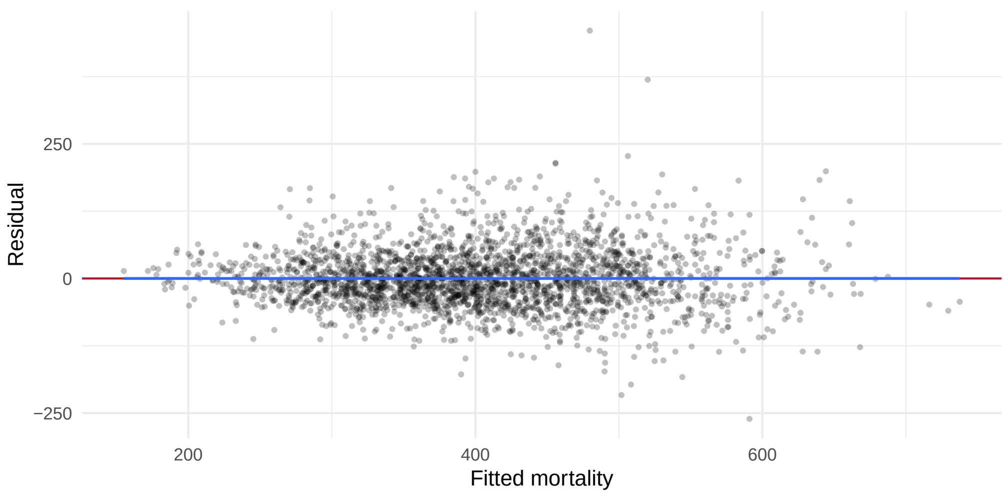

Figure 4: Residuals vs fitted values for the OLS model.

The residuals-versus-fitted plot (Figure 4) scatters around zero with no strong curvature, so the linear specification is reasonable. Their spread widens modestly at higher fitted values, indicating mild heteroskedasticity; this leaves the coefficient estimates unbiased but means their standard errors are best read as approximate, and a robust-standard-error or weighted specification would be a sensible robustness check.

6 Discussion and conclusion

Across more than 2,600 U.S. counties, premature mortality reflects several social conditions acting together. The multivariate model supports five of the six hypotheses: behavioural risk (smoking, obesity), limited healthcare access (uninsured), socioeconomic disadvantage (child poverty), and a harmful drug environment (opioid dispensing) are each independently associated with higher premature mortality (H1, H2, H3, H5, H6), and together they explain about 73% of the cross-county variation. This pattern is consistent with the social-determinants-of-health framework: place-based health is shaped jointly by how people behave, how disadvantaged they are, whether they can reach care, and the local drug environment, rather than by any single cause. H4 is the exception — graduation rate shows no protective *independent* effect once the other determinants are controlled — but, as argued above, this is consistent with education acting *through* behaviour and poverty, not with education being unimportant.

These findings also illustrate several broader methodological concepts. The graduation result shows why *confounding* and control variables matter: a bivariate relationship can change once related social conditions are held constant. The county-level design also creates an *ecological-inference* problem, so the coefficients describe places rather than individuals. Finally, the complete-case sample and mild heteroskedasticity show how sampling and model assumptions shape how confidently the results can be generalised.

The strongest caveat is design. Because the analysis is *ecological* and *cross-sectional*, associations measured across counties cannot be transferred to individuals — a county-level coefficient does not describe any one person's risk (Robinson, 1950) — and they identify association rather than causation.

Beyond design, the data impose further limits. The complete-case sample reflects listwise deletion, with missingness concentrated in the graduation-rate and opioid-dispensing variables, which may bias the sample toward counties with more complete reporting (typically larger, less rural ones). There is also a *temporal mismatch*: the opioid measure is from 2016 while the outcome is from 2018, which can be read benignly as a lagged exposure but remains a limitation. Finally, the model omits plausible *confounders* — county age structure, racial composition, rurality, healthcare infrastructure, and state-level policy — any of which could inflate or mask the estimated relationships.

The clearest next step is a spatial or multilevel model that tests the ecological gap rather than assuming it. State fixed effects would absorb regional clustering and unobserved state policy; an age-adjusted comparison would separate demographic structure from health behaviour; and multiple imputation with sensitivity checks would show how far the complete-case restriction shapes the results. Even within these limits, the analysis suggests that county premature mortality is a multi-determined outcome, in which supply-side drug factors and entrenched socioeconomic disadvantage both warrant policy attention.

7 Use of AI

Generative AI tools were used as a support aid: to help structure the report, to debug and tidy the R code, and to improve wording. All research-design decisions, variable selection, data handling, statistical interpretation, and final written content are my own and have been reviewed by me. AI was not used to fabricate data, results, or citations. References were reviewed for consistency and relevance before release.

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